Volume Breakout V2B

Cross-asset crypto strategy — BTC · ETH · SOL (4h)

Backtest 2018-01-01 → 2026-04-01 | $10,000 demo | Binance spot

Metric	Strategy	Buy & Hold
Final equity	**$250,760**	$80,423
Total return	**+2407.6%**	+704.2%
CAGR	**+47.8%**	+28.8%
Sharpe	**1.24**	0.71
Sortino	**1.08**	—
Max drawdown	**-31.0%**	-90.9%
Calmar	**1.54**	0.32

Equity multiple vs BH : 3.12× | DD reduction : 0.34× | Calmar ratio : 4.87×

Portfolio composition

BTCUSDT 60 % | ETHUSDT 25 % | SOLUSDT 15 %

Strategy parameters (walk-forward selected)

`don_len=30, vol_mult=1.3, regime_len=150, tsl_atr=4.5`

Walk-forward verification

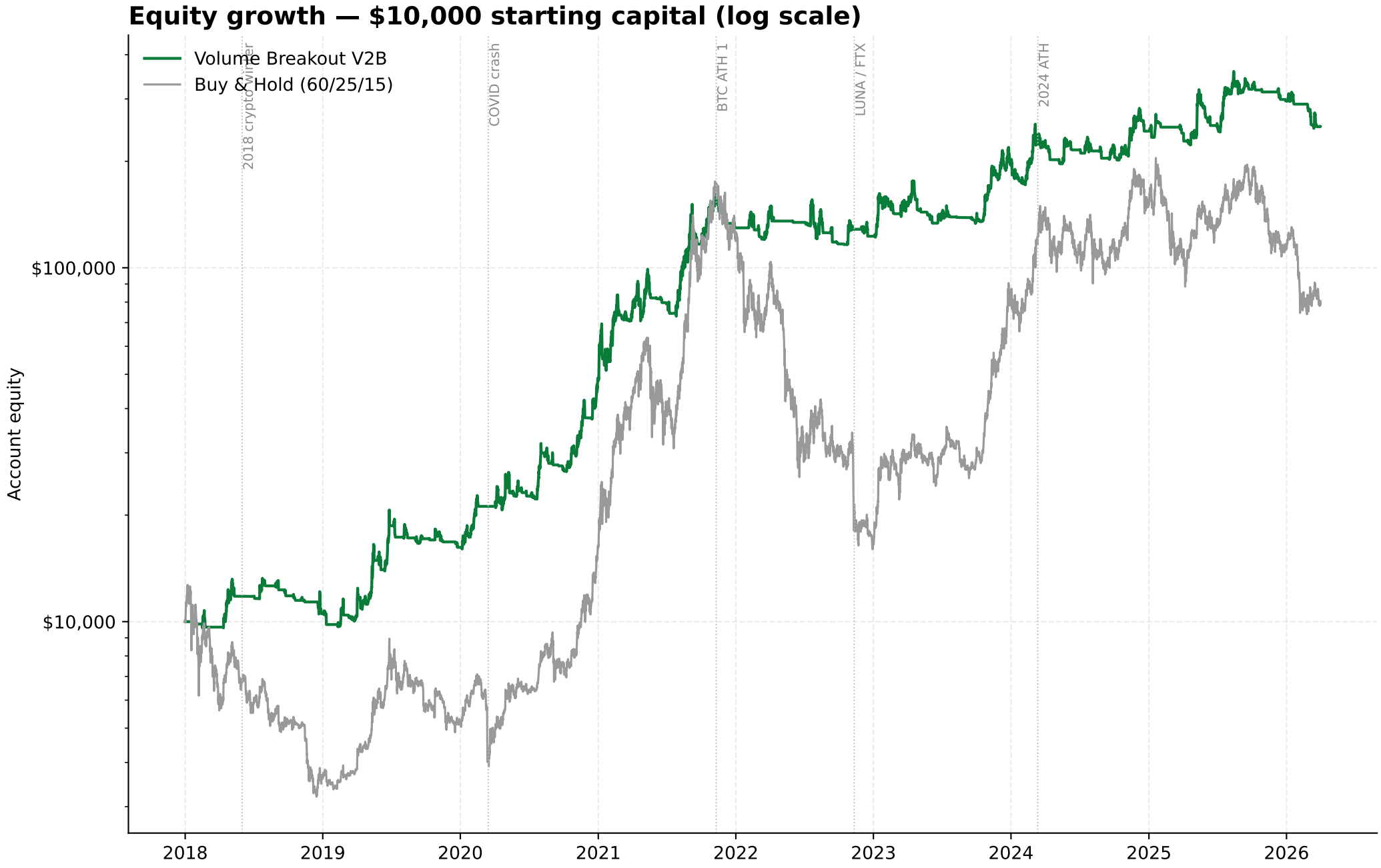
IS 2018-2022 : CAGR 65.2 % Sharpe 1.47 DD −27.9 %

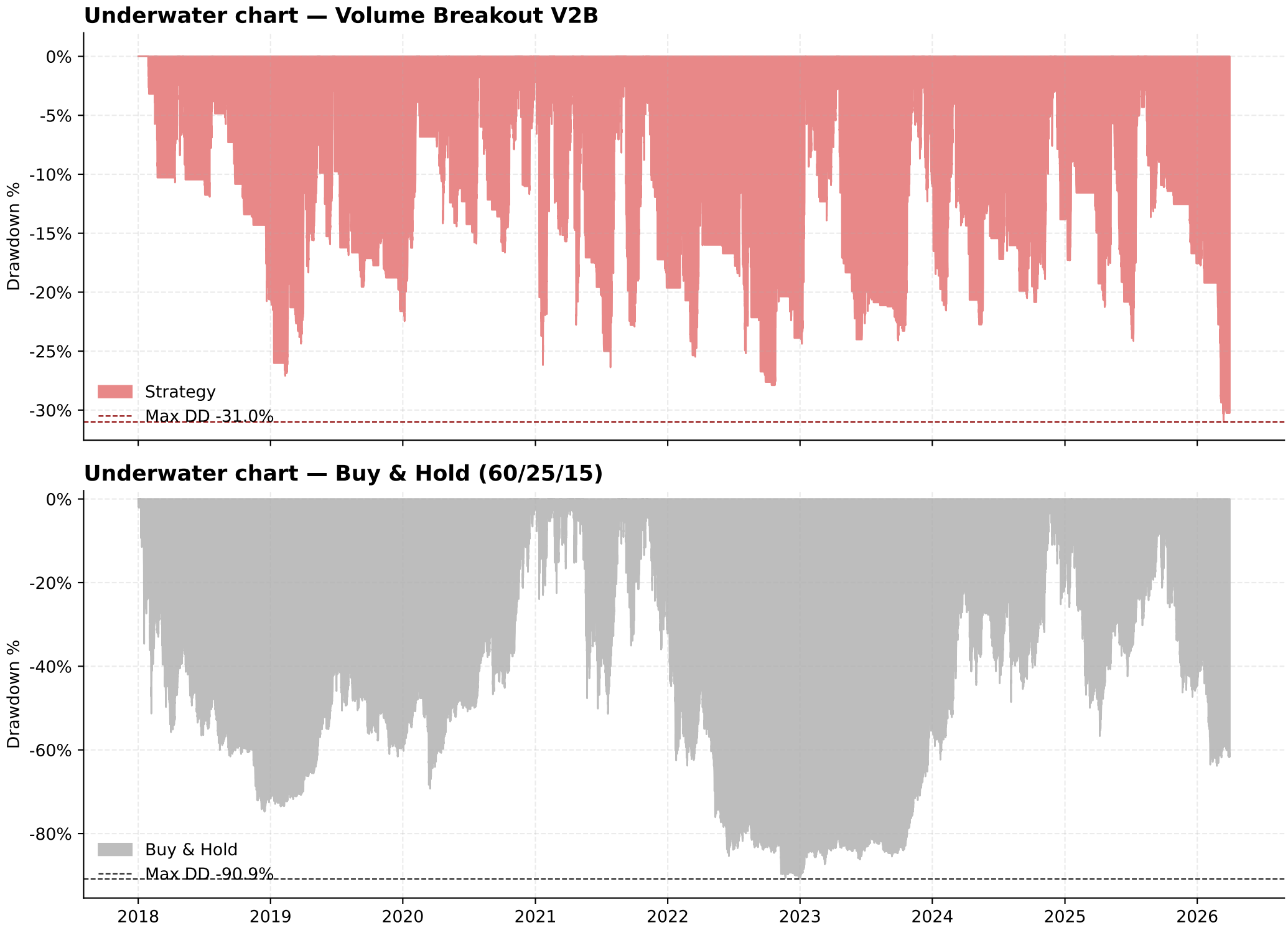
OOS 2023-2026 (unseen): CAGR 20.2 % Sharpe 0.79 DD −24.1 %

Per-asset contribution

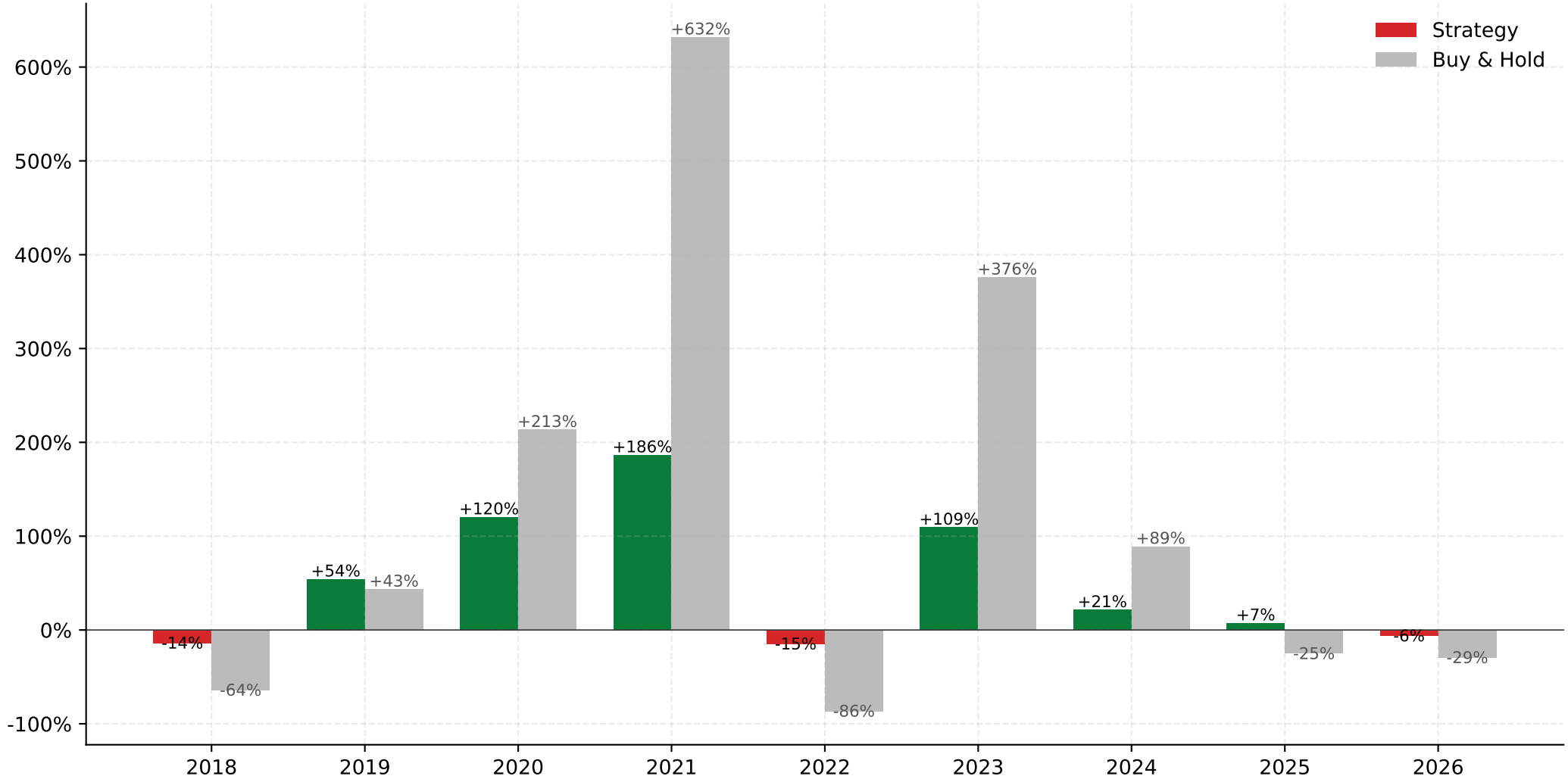
Asset	Init $	CAGR	Sharpe	MaxDD	Trades	Win %	BH return
BTCUSDT	$6,000	+24.3%	0.80	-49.2%	129	30.2%	+409.2%
ETHUSDT	$2,500	+62.5%	1.29	-37.6%	113	35.4%	+186.7%
SOLUSDT	$1,500	+101.6%	1.31	-71.9%	87	34.5%	+2747.0%

Costs modelled: 0.1 % per side + 5-tick slippage | Execution: next-bar OPEN after signal

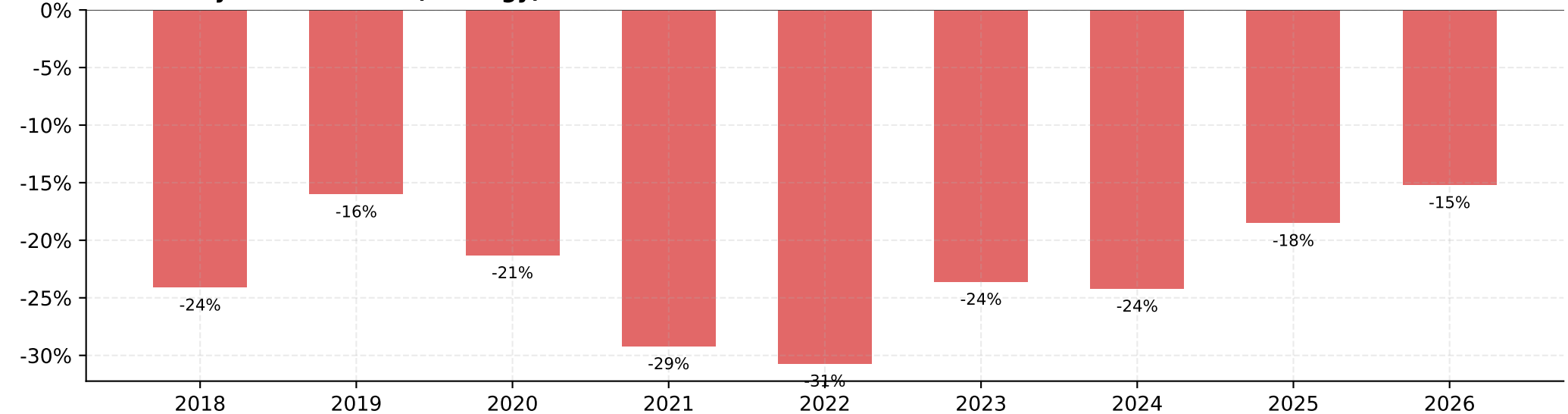

Equity growth — $10,000 starting capital (log scale)
Volume Breakout V2B
Buy & Hold (60/25/15)
2018 crypto winter
COVID crash
BTC ATH 1
LUNA / FTX
2024 ATH
$100,000
$10,000
Account equity
2018
2019
2020
2021
2022
2023
2024
2025
2026

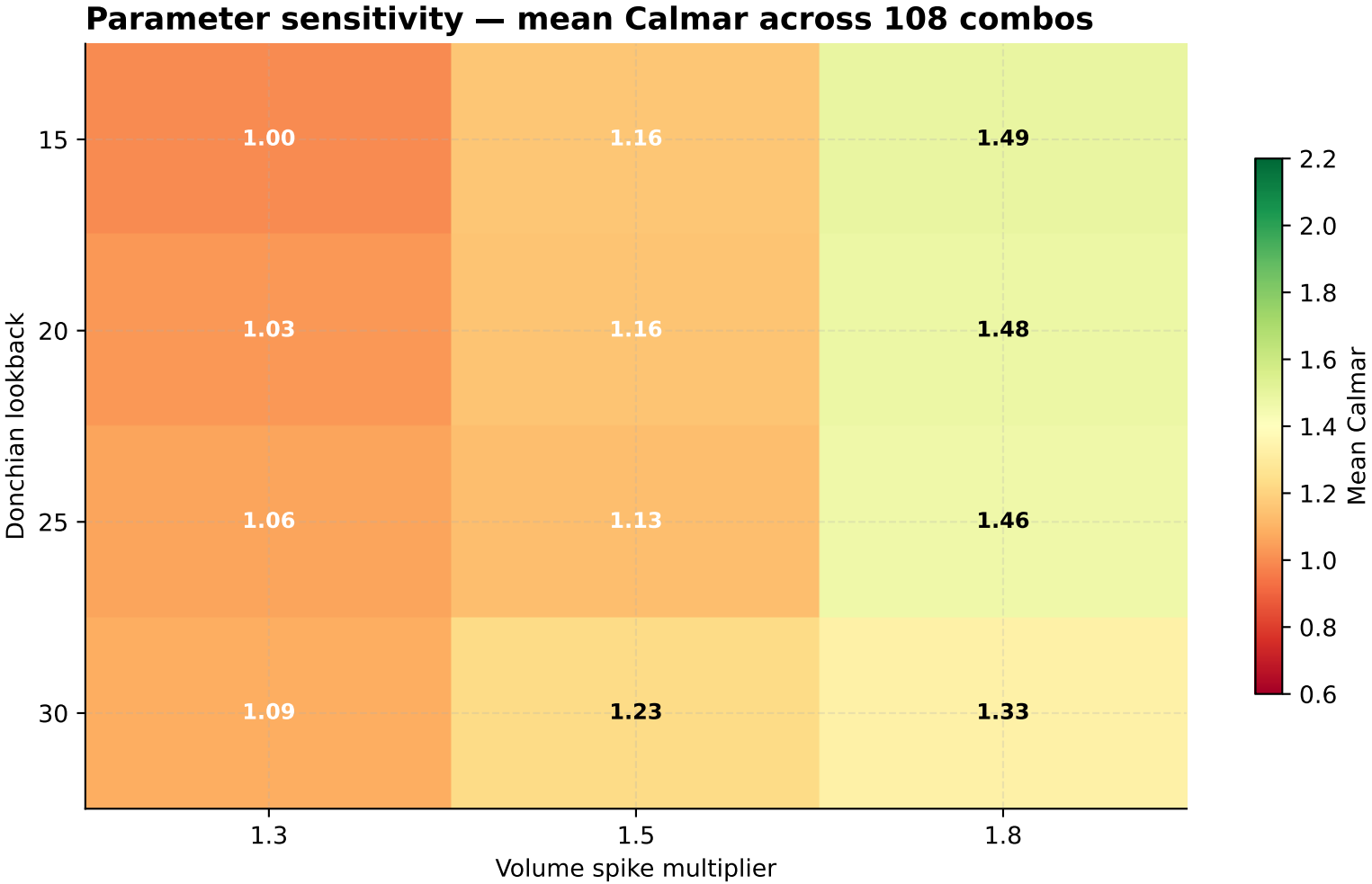

Underwater chart — Volume Breakout V2B
Drawdown %
0%
-5%
-10%
-15%
-20%
-25%
-30%
Strategy
Max DD -31.0%
Underwater chart — Buy & Hold (60/25/15)
Drawdown %
0%
-20%
-40%
-60%
-80%
Buy & Hold
Max DD -90.9%
2018
2019
2020
2021
2022
2023
2024
2025
2026

Annual returns — Strategy vs Buy & Hold



Max intra-year drawdown (strategy)





Stability across 108 combos: CAGR ∈ [22.1%, 55.1%] | Calmar ∈ [0.69, 2.11] | MaxDD ∈ [-45.6%, -22.3%] | 100 % of combos profitable

Rolling 1-year Sharpe ratio — strategy
90.7 % of rolling windows positive | 25th=0.46 75th=1.97
Rolling Sharpe
3
2
1
0
−1
Median 1.15
2019
2020
2021
2022
2023
2024
2025
2026

Strategy Logic

Entry (ALL three must be true on bar close):

```
1) close > highest(high, 30)[1]       breakout above prev 30-bar high
2) volume > sma(volume, 20) × 1.3     volume-spike confirmation
3) close > sma(close, 150)            higher-TF regime is up
```

Exits (whichever triggers first):

```
a) ATR(14) × 4.5 trailing stop        ratchets up from highest-since-entry
b) close < sma(close, 150)            regime failsafe
```

Execution & cost model:

- Signal on closed bar i -> FILL at OPEN of bar i+1 (no look-ahead)
- Fees : 0.1 % per side (Binance spot)
- Slippage : 5 ticks
- Position : 100 % of sub-account equity per trade, no pyramiding

Risk-adjusted portfolio allocation:

```
BTC 60 %   ETH 25 %   SOL 15 %   of $10,000 demo
Rationale: higher-quality / lower-vol assets get the largest share.
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Known limitations:

- Long-only. Shorts were tested and consistently hurt in crypto bulls.
- Single TF (4h). Adding multi-TF confirmation could improve OOS stability.
- SOL asset-level DD of -71.9 % — portfolio only averages to -31 % because of imperfect correlation between assets.
- OOS Sharpe degraded by ~46 % vs IS. Expect live Sharpe ~0.8-1.0.

Planned optimisation: add trend-validator filter

- HTF 1d regime overlay: only allow longs when 1d 200-EMA rising
- ADX(14) > 20 gate — avoid entries in chop
- Composite trend score (MA slope + ADX + rising-vol) with threshold
- Aim: keep CAGR, cut MaxDD below -25 %, lift OOS Sharpe > 1.0